

[0.25145, 0.07095, 0.00133, 0.0, 0.0, 0.0, 0.00106, 0.07194, 0.25077]

For $M = 2$, $\beta = 0.3$, $E_b N_0 \text{dB} = 15$:

Possible time intervals for sampling received signal: [-4 -3 -2 -1
0 1 2 3 4]

SER values for each time interval: [0.25145, 0.07095, 0.00133, 0.0, 0.0, 0.0, 0.00106, 0.07194, 0.25077]

The best time intervals for sampling the received signal are: [-1 0
1]

For $M = 2$, $\beta = 0.5$, $E_b N_0 \text{dB} = 15$:

Possible time intervals for sampling received signal: [-4 -3 -2 -1
0 1 2 3 4]

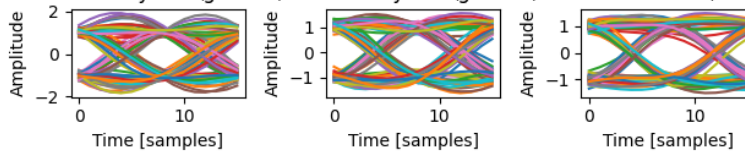
SER values for each time interval: [0.24958, 0.02765, 3e-05, 0.0, 0.0, 0.0, 1e-05, 0.02863, 0.24955]

The best time intervals for sampling the received signal are: [-1 0
1]

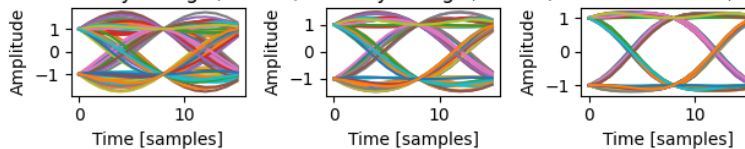
For $M = 2$, $\beta = 0.8$, $E_b N_0 \text{dB} = 15$:

Possible time intervals for sampling received signal: [-4 -3 -2 -1
0 1 2 3 4]
SER values for each time interval: [0.25338, 0.00265, 0.0, 0.0, 0.0
, 0.0, 0.0, 0.00246, 0.25024]
The best time intervals for sampling the received signal are: [-2 -1
0 1 2]

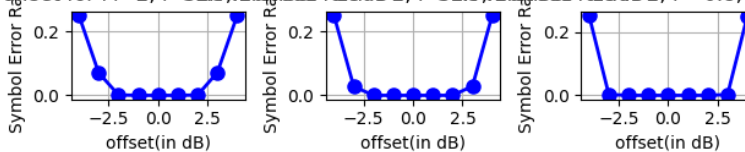
Eye Diagram with noise for $M=2$, $\hat{\rho}^2=0.8$, $E_bN_0dB=15dB$



Eye Diagram without noise for $M=2$, $\hat{\rho}^2=0.8$, $E_bN_0dB=15dB$



SER vs Offset for $M=2$, $\hat{\rho}^2=0.8$, $E_bN_0dB=15dB$



[0.56664, 0.43915, 0.23309, 0.01699, 0.0, 0.01719, 0.23178, 0.43947
, 0.56527]

For $M = 4$, $\beta = 0.3$, $E_bN_0dB = 15$:

Possible time intervals for sampling received signal: [-4 -3 -2 -1
0 1 2 3 4]
SER values for each time interval: [0.56664, 0.43915, 0.23309, 0.01
699, 0.0, 0.01719, 0.23178, 0.43947, 0.56527]

The best time intervals for sampling the received signal are:[0]

For $M = 4$, $\beta = 0.5$, $E_bN_0dB = 15$:

Possible time intervals for sampling received signal: [-4 -3 -2 -1
0 1 2 3 4]
SER values for each time interval: [0.56457, 0.42248, 0.18055, 0.00
429, 0.0, 0.00451, 0.17902, 0.42326, 0.56595]

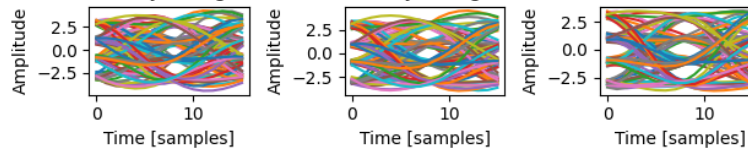
The best time intervals for sampling the received signal are:[0]

For $M = 4$, $\beta = 0.8$, $E_bN_0dB = 15$:

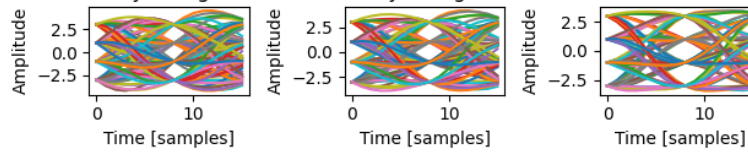
Possible time intervals for sampling received signal: [-4 -3 -2 -1
0 1 2 3 4]
SER values for each time interval: [0.56557, 0.3887, 0.09995, 0.000
36, 0.0, 0.00031, 0.099, 0.38804, 0.56568]

The best time intervals for sampling the received signal are:[0]

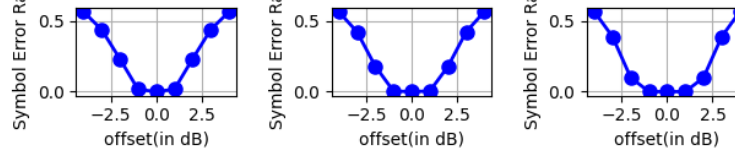
Eye Diagram with noise for $M=4$, $\hat{\tau}^2=0.8$, $E_bN_0dB=15dB$ Eye Diagram with noise for $M=4$, $\hat{\tau}^2=0.5$, $E_bN_0dB=15dB$ Eye Diagram with noise for $M=4$, $\hat{\tau}^2=0.1$, $E_bN_0dB=15dB$



Eye Diagram without noise for $M=4$, $\hat{\tau}^2=0.8$, $E_bN_0dB=15dB$ Eye Diagram without noise for $M=4$, $\hat{\tau}^2=0.5$, $E_bN_0dB=15dB$ Eye Diagram without noise for $M=4$, $\hat{\tau}^2=0.1$, $E_bN_0dB=15dB$



SER vs offset for $M=4$, $\hat{\tau}^2=0.8$, $E_bN_0dB=15dB$ SER vs offset for $M=4$, $\hat{\tau}^2=0.5$, $E_bN_0dB=15dB$ SER vs offset for $M=4$, $\hat{\tau}^2=0.1$, $E_bN_0dB=15dB$



In []:

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